

Problem L. Least Common Ancestor

OS Linux

Given an array of unique elements, construct a Binary Search Tree. Now, given two nodes u and v of the BST, find their Lowest Common Ancestor (LCA). LCA is defined as the furthest node from the root that is an ancestor for both u and v .

Input Format

First line of input contains T - number of test cases. First line of each test case contains N, Q - number of nodes in the BST and the number of queries. The next line contains N unique integers - value of the nodes. Its followed by Q lines, each contains 2 nodes of the tree, u and v .

Constraints

$1 \leq T \leq 1000$
 $1 \leq N, Q \leq 1000$
 $0 \leq ar[i] \leq 10000$

Output Format

For each test case, for each query print the LCA of the given nodes u and v , separated by space. Separate the output of different test cases with a newline.

Input	Output
2 5 2 3 2 4 1 5 2 5 1 2 7 3 4 5 15 0 1 7 17 0 15 7 17 17 4	3 2 4 15 4

Explanation o

Self Explanatory